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Perfusion Mammalian Cell Culture for Recombinant Protein Manufacturing

In this introductory chapter, we discuss mammalian cell culture for the production of therapeutic proteins in the broader context of biotechnology and in particular of the biopharmaceutical industry. We begin with a short retrospect on the history of cell cultures for bioproduct manufacturing and eventually introduce recombinant technology, in order to appreciate how the present standards were established. An overview of the current market on recombinant therapeutic proteins provides some important understanding of the industry challenges to come and the contribution that continuous manufacturing can provide. We then introduce the various bioreactor types that can be used in this area and indicate the challenges that are faced in their development, design and operation.

The objective here is to put all these aspects in the right perspective and direct the reader to the chapters where each of them is specifically treated this book.

1.1 BIOTECHNOLOGY: FROM EARLY APPLICATIONS TO BIOTHERAPEUTICS

In general, biotechnology refers to any activity that uses living systems to create value through services or goods. The immense diversity of applications increases continuously with new discoveries in fundamental biology and, more recently, also with advances in computational sciences that allow extracting new information from the incredible amount of data generated with increasingly sophisticated analytical techniques.

Therapeutic proteins constitute a particularly important and lively part of this discipline. The number of indications treated with therapeutic proteins is increasing at a fast pace, which is very encouraging for patients suffering from various diseases – in particular, of autoimmune and oncologic nature. On the other hand, this represents an enormous market with sales in the order of billions of dollars each year, where manufacturing technologies play a fundamental role (Ecker et al. 2015, Walsh 2014, Zhang et al. 2016). Fermentation is probably one of the most common of such technologies and appears in a variety of complex industrial applications. It is probably more convenient to approach its description from an historical prospective, which is rooted in food processing.

1.1.1 First Applications of Fermentation

Humans began exploring biotechnology thousands of years ago, long before the existence of microorganism, genes or deoxyribonucleic acid (DNA) could even

be imagined. However, progress in the area, as it is often the case, did not proceed continuously but through distinct discontinuities. Sometimes knowledge was stagnant for decades or even centuries until some technological breakthrough moved it to the next phase, through a sequence of events eventually leading to the modern context of biotechnology and biomanufacturing. The first and by far the longest phase started with food and beverage fermentations using mixed cultures of organisms whose identity remained unknown at least until the nineteenth century. Their primary function was to improve the preservation of aliments but eventually turned into the production of more sophisticated products. Cheese-making began about 8,000 years ago using microorganisms such as bacteria, yeasts and fungi to transform milk into a wonderfully flavoured but, most importantly, conservable product (Beresford et al. 2001). Fermentations of meat and vegetables also started early on and are still in use today not only for conservation purposes but also to create flavours that are part of the culinary legacy of our ancestors.

Fermented beverages could be traced back even earlier to the seventh millennia B.C. (McGovern et al. 2004). Wine and beer are part of the most ancient fermentation processes and are still today produced in large amounts, with all the positive and negative implications this has had on societies ever since. The oldest evidence of advanced beer brewing was traced back to around 5,000 years ago in China (Wang et al. 2016), while evidence for wine fermentation is traced to about 6,000 years ago, in today's Iran (McGovern et al. 1996, Soleas et al. 1997). These processes are currently well characterised, and methods for continuous beer production are being developed (Brányik et al. 2005, Pilkington et al. 1998).

It was only during the nineteenth century that fermentation was correlated to the activity of living organisms (Manchester 2007, Pasteur 1885). During the industrial revolution and with the advances in microbiology, these processes started to be understood and controlled (Caplice & Fitzgerald 1999).

1.1.2 Extraction of Valuable Metabolites and Proteins

The next technological milestone was related to purified monocultures of selected microorganisms used for the production of primary metabolites. The pioneer of these processes was probably Chaim Weizmann with his studies on the fermentation of acetone, butanol and ethanol in the 1910s. World War I motivated the development of fermentation technologies because of the large use of acetone and glycerol in munition production (Demain 2007, Wang et al. 2001). Many products were then synthesised using microorganisms including amino acids, vitamins, nucleotides, solvents and organic acids (Demain 2007, D'Este et al. 2017, Sano 2009, Stahmann et al. 2000, Zhang et al. 2016). The application of fermentation for the production of secondary metabolites can be regarded as a new technological breakthrough – which allowed the production of a new series of products such as toxins, pesticides and animal or plant growth factors and, of course, antibiotics, which alone had an unmeasurable impact on mankind since their discovery in 1929 by Alexander Fleming (Fleming 1929).

1.1.3 Recombinant DNA and the Manufacturing of Recombinant Proteins

Until this point, biotechnology was using existing organisms as such, in a controlled environment, with the purpose of collecting some selected secreted components.

1.1 Biotechnology: From Early Applications to Biotherapeutics

Cultivation methods for industrialisation were developed mainly with microbial systems. The next and probably most significant breakthrough came through the so-called recombinant DNA technology developed after some pioneering work of three scientists – Paul Berg, Herbert Boyer and Stanley Cohen – in the 1970s. Actually, it was not so much the technique itself that was revolutionary but, rather, its application, which changed biology from a largely analytical science into a synthetic one (Berg et al. 2007). At that time, the access to therapeutic proteins was extremely limited and of animal origin. For example, insulin was of bovine or porcine origin and had significant adverse effects on the patients. With recombinant technologies, the human form of insulin could be produced at much higher safety and efficiency levels. The possibility to synthesise, multiply and integrate the DNA in a different expression system was revolutionary and had a terrific impact on medicine and many fields in biology, to the extent that it even changed the fundamental concepts of intellectual property for biological inventions (Berg & Mertz 2010, Cohen et al. 1973, Hughes 2001). Figure 1.1 shows Herbert Boyer on the *Time* magazine cover (next to one of the first appearances of the future Princess Diana). He co-founded Genentech (standing for **GEN**etic **ENG**ineering **TECH**nologies); which, even today is one of the major companies in the biopharma world. The development of the technology for synthesising recombinant DNA is summarised through the major corresponding milestones in Table 1.1.

Recombinant technology is used for the production of highly specific therapeutic proteins. The DNA sequence coding for the protein of interest is inserted into a living host that is able to translate the DNA sequence into an amino acid sequence. When dividing, the host propagates the DNA information, or sequence, and synthesises the protein using its innate machinery (Colosimo et al. 2000). As mentioned earlier, microorganisms such as bacteria, yeasts, fungi, insects, plants and mammalian cells were used for a long time in this industry, but the development of recombinant DNA technologies opened the doors to entirely new production horizons. Indeed, even today, only the complex machinery of a living organism is able to synthesise large proteins to achieve desired biological activities. However, we now have the choice of which host to use, and this is often a key and decisive question. Evolution offered us a diversity of organisms that all have the same DNA coding system for the assembly of amino acids but differ in terms of growth behaviour, post-translational modification (PTM) capabilities, secretion efficiency, etc. Indeed, after transcription (DNA into RNA) and translation (RNA into an amino acid sequence), the cell machinery takes care of the protein folding, the addition of different sugar molecules (glycosylation) and several other reactions known as PTMs (Beyer et al. 2018, Higel et al. 2016, Walsh 2010). These changes are regulated by a complex network of enzymes and take place in the endoplasmic reticulum (ER) and in the golgi apparatus, as schematically illustrated in Figure 1.2. As an example, glycosylation can be described as a sequential processing of an oligosaccharide that is added to the protein structure in the ER. These added glycans are known to impact not only the stability but also the biological activity of the protein in the human body (Frenzel et al. 2013; Hossler et al. 2009; Jefferis 2005, 2009). A schematic view on how these transformations develop is shown in Figure 1.3. Starting from the upper left part, it is seen that the oligosaccharide chain is attached to the protein backbone in the ER. Then, some enzymatic reactions transform this glycan into oligomannose form, which is transferred to the Golgi apparatus – where, through other enzymatic reactions, different saccarides are removed or added. This complex

Table 1.1 Major events in the 1970s that are related to the development of recombinant DNA production and cloning methods © Republished with permission of Genetics Society of America, from personal reflections on the origins and emergence of recombinant DNA technology', P. Berg and J. Mertz, 184, 2010; permission conveyed through Copyright Clearance Center, Inc.

Year	Event
1969–1970	P. Berg and Lobban independently conceive ideas for generating recombinant DNAs <i>in vitro</i> and using them for cloning, propagating and expressing genes across species.
1971	Berg (1974) isolate the first plasmid bacterial cloning vector, <i>dvgal</i> 120.
1971	Robert Pollack first raise the concern regarding potential biohazards of cloning.
1971–1972	Jackson et al. (1972) and Lobban and Kaiser (Lobban 1972, Lobban & Kaiser 1973) concurrently and collaboratively develop the terminal transferase tailing method for joining together DNAs <i>in vitro</i> .
1972	Jackson et al. (1972) create first chimeric DNA <i>in vitro</i> .
1972	Mertz and Davis (1972) discover that cleavage with EcoRI generates cohesive ends. They use EcoRI plus DNA ligase to generate SV40 – <i>dvgal</i> 120 chimeric DNAs <i>in vitro</i> .
1972–1973	Cohen et al. (1972) isolate the drug-scalable bacterial cloning vector, psC101. They use it to construct, clone and express bacterial intra- (1973) and inter-species recombinant DNAs.
1973	Morrow et al. (1974) clone and propagate ribosomal DNA genes from a eukaryote in <i>E. coli</i> .
1973–1976	Renewed concerns regarding potential biohazards of cloning recombinant DNAs (Berg 1974, Berg et al. 1975, Singer & Soll 1973) lead to NIH Guidelines.
1974–1975	The initial Standford University/University of California, San Francisco (UCSF) (Cohen/Boyer) patent application relating to recombinant DNA is filed.
1976	Boyer and Swanson co-found Genentech, the first biotechnology company.
1980	Stanford/UCSF (Cohen/Boyer) patent issued by U.S. Patent Office.

reaction network can lead to a wide variety of different glycan species, which can be identified as more or less specific structures. It is very common to distinguish among different galactosylated forms, which include glycans that have one or more galactose attached. In this book, we distinguish among groups that are not galactosylated (G0) or have one (G1) or two (G2) molecules of galactose attached, as well as high mannose (HM) forms, referring to molecules exhibiting five or more mannose.

All these modifications play a fundamental role in the protein functionality and stability and are therefore of the greatest importance. This is the reason why mammalian cells are widely used in the production of therapeutic proteins, and this is the host we will focus on in the following. Although their cultivation is more complex and less effective in terms of productivity compared to bacterial or even yeast systems, their capacity to modulate certain post-translational modifications – like glycosylation, for example – makes them a powerful host for the production of recombinant proteins that are biologically active and safe for humans. In the next chapter, we discuss existing cultivation methods for mammalian cells.

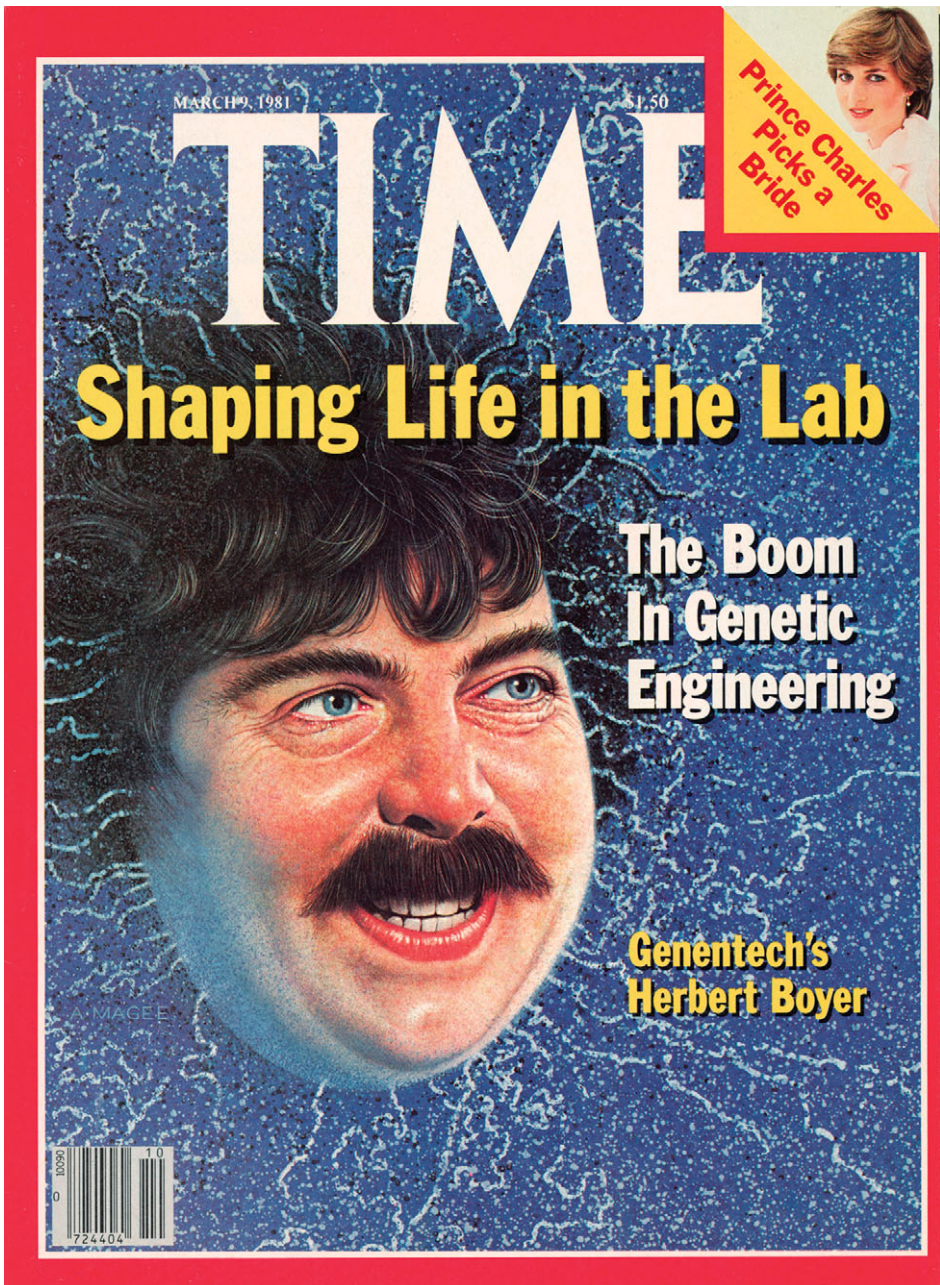


Figure 1.1 *Time* magazine cover (1981) showing Herbert Boyer, Professor of Microbiology and Biochemistry at the University of California in San Francisco and co-founder of Genentech (Hughes 2001). From TIME. © 1981, TIME USA LLC. All rights reserved. Used under licence

1.1.4 Product Quality and Quality by Design

Similar to any other industrial product, the quality of therapeutic proteins can be assessed through a list of physico-chemical characteristics, referred to as critical quality attributes, that have to fall within strict specifications. These are defined and

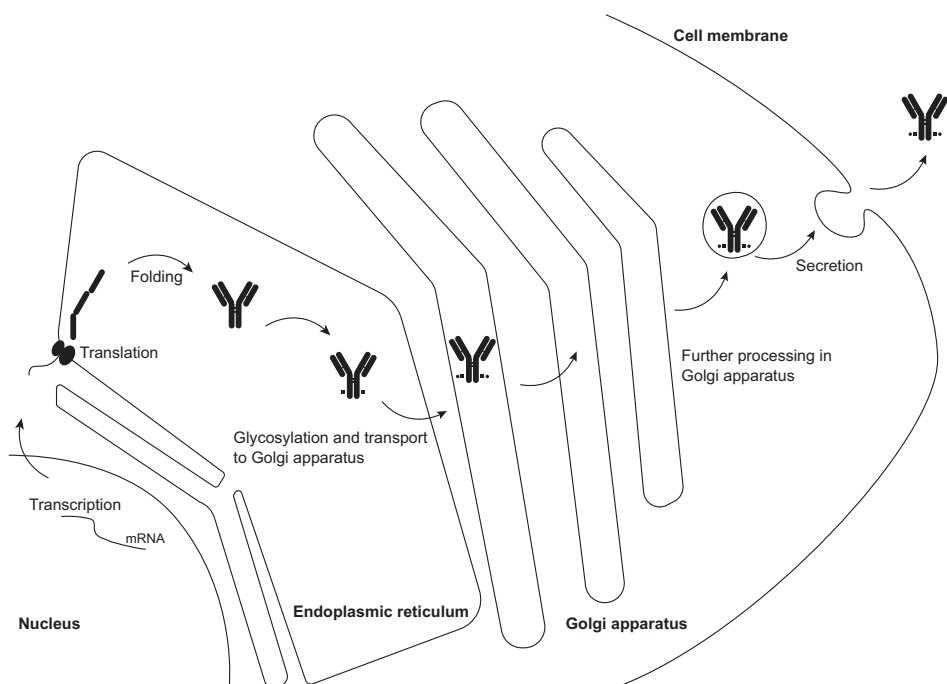


Figure 1.2 Protein pathway through the cell before secretion. After translation, the protein is folded in the ER, and some post-translational changes start to happen. The protein then travels through the Golgi apparatus, where further glycosylation transformations are regulated. The protein is then secreted to its functional location (Hu 2012).

controlled by the regulatory authorities in order to guarantee safety and efficacy of the drug. Opposite to chemically synthesised small-molecule drugs, which can be univocally defined by a single molecular structure, biological products, because of their size and complexity, are actually constituted by a heterogeneous population of very similar variants or isoforms. The sequence of assembled amino acids that represents the protein primary structure is controlled, but the host cell line and the process to express the protein have a direct impact on the PTMs – such as glycosylation or undesired chemical transformations like deamidations, oxidations or others which lead to undesired species or impurities. A biological product is therefore defined also by the corresponding production process, and its structure is defined based on the distribution of the different quality attributes. This is done through acceptable ranges that are more or less narrow depending on each attribute. One of the consequences is the difficulty of stating that two bioproducts are identical, which makes it intrinsically impossible to copy an existing biological drug. For this, the notion of generic that is used in the small-molecule pharma industry does not apply to the biopharma world. The concept of biosimilar was therefore developed to cope with the heterogeneity of the products in this industry. Guidelines were given by the authorities to define biosimilars and help manufacturers developing strategies for their production (Brühlmann et al. 2015).

In addition to the heterogeneity mentioned earlier, critical quality attributes (CQAs) include impurities which derive from the protein production process. Examples include DNA and host cell proteins (HCP) coming from the metabolism

1.1 Biotechnology: From Early Applications to Biotherapeutics

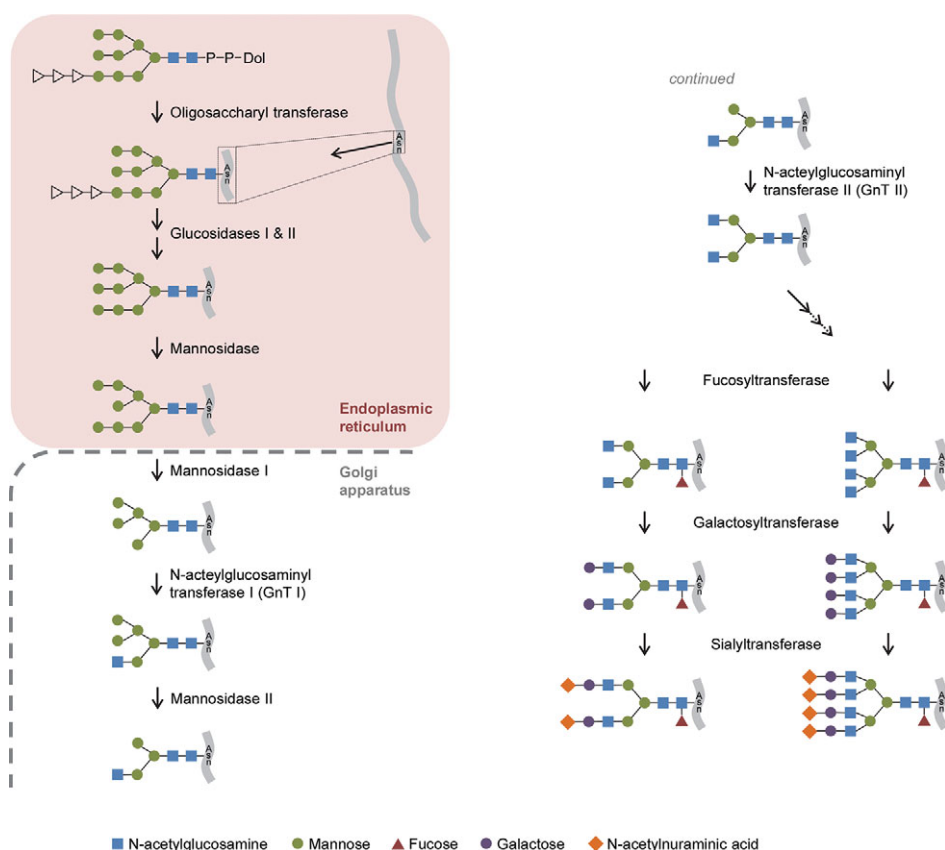


Figure 1.3 Schematic of glycosylation patterns in the ER and the Golgi apparatus. © Reprinted from *Biotechnology Progress*, David Brühlmann, Martin Jordan, Jürgen Hemberger, Markus Sauer, Matthieu Stettler and Hervé Broly, 31/3, 'Tailoring recombinant protein quality by rational media design', 2015, with permission from John Wiley and Sons.

of the host cells or low molecular weight (LMW) species and high molecular weight (HMW) species forms arising from protein fragmentation and aggregation processes, respectively. In general, all impurities are classified as product or process related, as shown in Table 1.2, and need to be properly taken care of in either the upstream or the downstream part of the production process. This requires a delicate coordination between the two because, as we will discuss later, it may be easier, if at all possible, to satisfy specifications for one specific critical quality attribute in one or the other of the two parts of the process.

The heterogeneity and variability of these attributes depends, on the one hand, on the given expression system and protein of interest, but also on the process used for the production of the molecule. Within product quality considerations, the concept of quality by design (QbD) has been strongly promoted in recent years (Woodcock 2014). QbD proposes a shift towards a better understanding of the product, its manufacturing process and the identification of potential critical product quality attributes (Rathore & Winkle 2009, Yu et al. 2015). The goal of QbD is to understand how these parameters are related to the process parameters and to develop quantitative relations among the two so as to design product quality based on process operating

Table 1.2 Non-exhaustive list and description of some typical product CQAs and some examples of CPPs that could affect them.

Critical quality attribute	Critical process parameter	Type	Description
Host cell proteins	Viability and cell density	Process	Impurity that may trigger immunogenic responses.
DNA	Viability and cell density	Process	Impurity that may trigger immunogenic responses.
Aggregates (HMW)	Titre and process duration	Product	Impurity that leads to loss of function and potentially in immunogenic responses.
Fragments (LMW)	Titre and viability	Product	Impurity resulting from chemical or proteolytic degradation; it can lead to loss of function and potentially immunogenic responses.
Charge variant	pH and temperature	Product	The charge distribution on a protein is often heterogeneous and impacts structure, stability, binding affinity, etc. Chemical degradations can modify the charge distribution and generate acidic or basic variants.
Glycosylation	pH and metabolite concentrations	Product	The glycoform distribution is characterised by various lumped forms: G0, G1, G2, sialylation, fucosylation, high mannose forms etc. Their proportions impact the biologic activity of the molecule.

conditions. The objective is to control, through the product quality attributes, biological activity, pharmaco-kinetics and pharmacodynamics, as well as safety and toxicity, in order to produce consistently safe and therapeutically efficient drugs (Rathore 2009, Rathore & Winkle 2009, Rouiller et al. 2012).

The quality of the product is, of course, strongly affected by the manufacturing process, both upstream and downstream. Critical process parameters (CPPs) are identified as acting on CQAs and must therefore be closely monitored. Downstream processing is typically regarded as the step where some undesired features of the product leaving the bioreactor can be modified to satisfy specifications, but this is not always possible. Indeed, downstream processes, whether in batch or continuous mode, are capable to remove or reduce within specifications process-related impurities, such as DNA and host cell proteins, as well as product-related impurities, such as aggregates and fragments (Steinebach et al. 2016b, 2017). Although more difficult, downstream processes – in particular, the multicolumn countercurrent solvent gradient purification (MCSGP) process – can change the charge variant patterns or isolate the main isoform at high purity and yield (Krättli et al. 2013). N-linked glycosylation isoforms, however, cannot be separated within the current downstream processes. Therefore, if needed, these patterns have to be modulated in the upstream process.

The use of continuous manufacturing, such as perfusion cell culture, impact the quality of the product in comparison to batch-like processes. This has relevant

implications also at the monitoring level, for which regulatory environments already exist (Allison et al. 2015).

1.2 MAMMALIAN CELL CULTURES FOR RECOMBINANT PROTEINS

Mammalian cells naturally grow in the adherent state, and therefore, their cultivation requires suitable carriers. Adherent cells are still used in some well-established manufacturing processes, but growing cells in suspension, obtained through proper adaptation procedures, is much more efficient and is currently the industrial standard. For a given bioreactor, the density reached with suspended cells is, in fact, much higher than with adherent cells simply because there are no surface limitations.

Interestingly though, in the 1990s, perfusion processes were developed using adherent cells. Examples of commercial products include the recombinant follicle stimulating hormone (Gonal-f ©), interferon beta-1a (Rebif ©) and factor-VIII (Kogenate-FS ©), which are all large and labile molecules (Pollock et al. 2013). For this reason, they were cultivated in perfusion, and not batch or fed-batch mode, since labile molecules require low bioreactor residence times. In addition, adherent cells are intrinsically compatible with simple and efficient cell retention. In particular, with adherent systems, either fixed carriers, which prevented the cells from being harvested, or microcarriers, constituted by small floating beads of polymer whose density could be exploited in sedimentation devices to retain them inside the vessel, were used. The rare cells or cell debris present in the harvest stream were simply removed using depth filtration. In the case of cell suspension systems, only very low cell densities could be achieved, thus making cell settlers followed by depth filtration sufficient for reaching satisfactory cell retention inside the perfusion bioreactor.

However, at that time, industry largely favoured batch-like operations. From the 1990s to today, in fact, technologies around expression systems, media and process control went through enormous improvements (Chu & Robinson 2001, Lim et al. 2010). The adaptation of cell lines to suspension cultures offered new possibilities for manufacturing sufficiently stable proteins. Batch operations with suspension cultures were favoured in industry because of the increased productivity potential related to the higher cell concentration that could be reached compared to adherent systems. Reliable retention devices for even modestly concentrated suspension cultures were not available at the time. From the 1990s to today, the productivity or yield of this type of processes doubled every five years (Langer 2014, Meuwly et al. 2006). On the other hand, continuous cell culture was also used, but with some limitations. Low cell densities were achieved in perfusion due to the surface limitation for adherent cells, or to the retention capacity for suspended cells. Also, the number of stable molecules such as monoclonal antibodies (mAbs) increased in the company pipelines. This also motivated the development of batch systems, since low residence time was not a priority anymore for these molecules. The consequence is that today the industry standard for commercial manufacturing is fed-batch in stirred tank reactors of up to 25 kL (Moyle 2017). These reactors offer large mixing and mass transfer rates and can be used to cultivate many different cell types using different operation modes (Rodrigues et al. 2010). The downside is the large capital investment they require – which, according to the Biophorum Operations Group (BPOG), amounts to up to hundreds of millions dollars at commercial scale for a line of four bioreactors (Sawyer et al. 2017a).

Current market trends have also been well summarised by the BPOG, and four major drivers have been identified the continuous growth of the biopharmaceutical market, the introduction of new product classes, the cost pressure coming from the payers and, finally, the uncertainty of approvals for the new products. These factors put leaders in a delicate situation in which not only uncertainties about sales but also manufacturing capacity needs are growing. The past decades were mainly dominated by some blockbusters for each company that were driving the capacity needs. With the coming changes, new strategies that require more flexibility are needed. Large stainless steel plants might not be the desired standard anymore, mainly due to the large capital investments and the long time (up to five years) required to be operational.

Continuous manufacturing is envisioned as one way to respond to the needs of these new market trends. If the production can be intensified, the bioreactor size, and its corresponding footprint, could be significantly reduced. Enabling technologies exist today for high cell density (up to 100×10^6 cells/mL and more) culture filtration with efficient cell retention (Barrett et al. 2018). This, together with the reduced lead time needed between the decision to build a commercial capacity and its first operation, promotes the introduction of continuous technologies. Single-use technologies are often mentioned in this context because they also offer an increased flexibility and a decreased capital investment.

In the following section, we briefly summarise the key aspects of the various bioreactor operating modes in order to put in a better prospective their capabilities and limitations.

1.2.1 Batch Operation

Batch operation is the simplest operation mode since it does not require any addition or removal of media during operation (Al-Rubeai 2015, Hu 2012, Zhou & Kantardjieff 2014). The bioreactor is inoculated at a defined cell density, and then the cells grow, while the product is being secreted, up to some maximum level. Parameters like temperature, pH, mixing and dissolved oxygen are controlled in the culture vessel. The media used to start the batch bioreactor contains enough nutrients to sustain the culture for the entire batch time. Components typically include amino acids, vitamins, trace elements, glucose, growth factors and salts. In the earlier times, these culture media also contained some supplements extracted from living beings such as serum of animal origin or peptones issued from yeasts or plants. These supplements were critical for safety and consistency reasons (lot-to-lot variations); therefore, media containing only known concentrations of defined components – usually referred to as chemically defined (CD) media – were developed. In all these cases, the media are constituted by a rather complex mixture of many components. In batch mode, the nutrient content is limiting and limited – limiting because once the nutrients are consumed, the cell culture is not sustained anymore, and limited not only because of solubility or stability constraints of each component but also because the starting condition must be in an acceptable pH and osmolality range in which the cells can grow. This implies that the concentrations cannot simply be increased indefinitely to increase the media capacity to grow cells.

Another limiting factor for the cell culture is the accumulation of toxic metabolites. During the culture, the cells secrete a diversity of metabolites that are known to be detrimental to cell health if too concentrated. Some of these metabolites like

1.2 Mammalian Cell Cultures for Recombinant Proteins

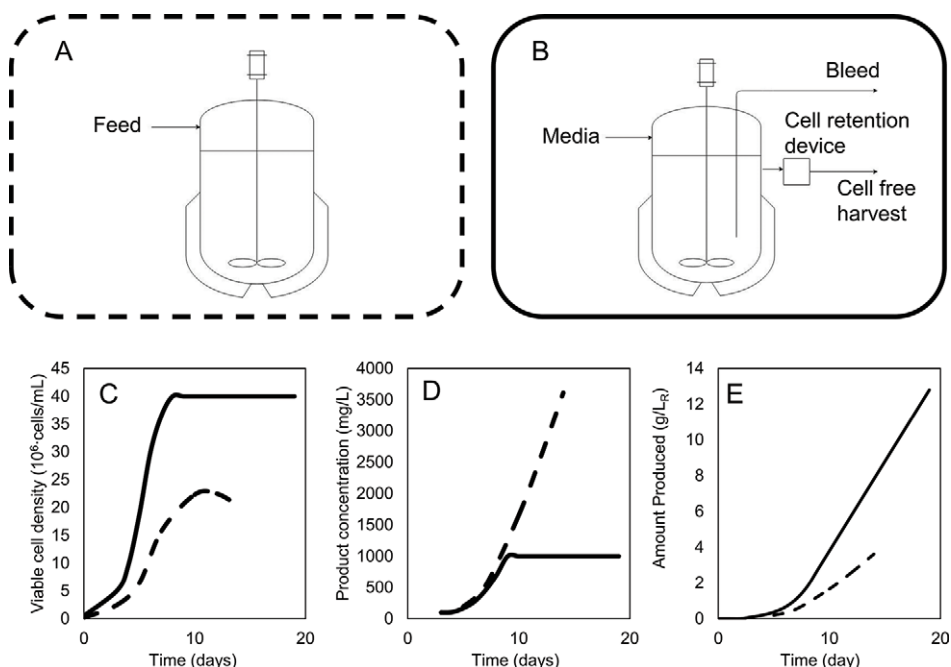


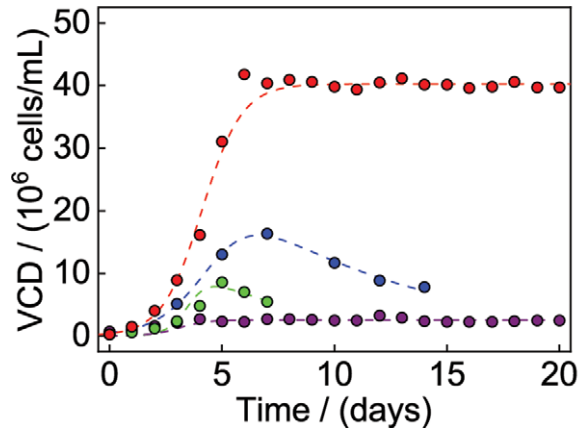
Figure 1.4 Schematic of fed-batch (A) and perfusion (B) culture of a recombinant cell line. In a fed-batch bioreactor, feeds are added during the run, and the product accumulates in the bioreactor. In perfusion, media is continuously fed, and harvest is continuously removed. The cell retention device clarifies the harvest stream, and the product is (in this example) not retained and, therefore, not accumulating. The bleed stream is used to remove excess cells and keep a steady concentration, in this example, at 40×10^6 cells/mL. The schematic depicts cell density (C), protein concentration inside the bioreactor (D) and cumulative specific production (E) as a function of time; broken lines represent fed-batch and solid lines perfusion. © Reprinted from *Biotechnology Advances*, 36/4, Jean-Marc Bielser, Moritz Wolf, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, 'Perfusion mammalian cell culture for recombinant protein manufacturing: A critical review', 2018, with permission from Elsevier.

ammonium or lactate are well known and can be monitored, but others, not well characterised, may also interfere with the culture.

1.2.2 Fed-Batch Operation

The operation of a fed-batch process is similar to a batch operation, but concentrated nutrient solutions (feeds) are added to the culture during operation to avoid the depletion of essential nutrients, as shown in Figure 1.4. This strategy enables the increase of the maximum cell density and also the culture duration with respect to batch operations. Various feed addition strategies are possible: it can be bolus or continuous addition, and the added volumes can be based either on an established platform or on some reference nutrient concentration. Several different feeds are often used, for example, to control the glucose concentration or avoid the depletion of specific nutrients. Some components that are unstable or difficult to solubilise might also require an additional feed stream. However, one issue is not resolved by this strategy: the accumulation of waste and possibly toxic metabolites in the bioreactor, which imposes in all cases an upper bound to the maximum achievable viable cell density.

Figure 1.5 Viable cell density as a function of time for batch (green), fed-batch (blue), chemostat (purple) and perfusion (red) bioreactors.



I.2.3 Perfusion Operation

Continuous cell culture operation is possible with a constant media addition – which, in order to keep constant the bioreactor volume, has to be coupled with a constant harvest flow. If no device is used to retain the cells in the bioreactor, this operation is referred to as chemostat. However, since the growth kinetics of the cell culture – especially with mammalian cells – is rather slow, it is preferred to retain the cells inside the bioreactor, which corresponds to the perfusion operation mode. One exception that needs to be mentioned here is the case where the separation device also retains the product of interest inside the bioreactor. This is a relatively rare situation which is not further considered in the following paragraphs (Yang et al. 2016).

A perfusion bioreactor is characterised by two outlet streams as shown in Figure 1.4: the harvest, which contains no cell and controls the concentration and residence time of the secreted product, and the bleed (containing also the cells), which is used to control the cell density in the bioreactor and is obviously significantly smaller. In this operation mode, the toxic metabolites are removed from the bioreactor while cells are retained, thus enabling maximum viable cell density values larger than in all other operations to be reached. This is illustrated in Figure 1.5, where the four operating modes are compared for a representative case. On the other hand, the nutrients and secreted product are also removed from the reactor, thus leading to lower titres and higher media-specific consumption. The various aspects relative to the operation, monitoring and control of perfusion bioreactors are discussed in Chapter 2.

Perfusion operation, as well as the chemostat, differs significantly from batch and fed-batch because they are open or continuous systems, with constant inlet and outlet flows, which after a sufficient transient time tend to balance each other and reach a time-independent state usually referred to as steady state. This is well described by the classical ideal model usually referred to as CSTR (Fogler 2008). A qualitative comparison of fed-batch and perfusion operations is shown in Figure 1.4C, D and E with respect to viable cell density, titre and cumulative production.

Because the maximum viable cell density that can be reached using perfusion is significantly higher than in batch or fed-batch operations, the limitations in terms of cell density and culture duration shift from nutrient depletion or toxic

1.3 Development and Operation of Perfusion Cultivations

compounds accumulation to cell retention, oxygen supply and cell stability. The challenges become to control the flow of the different streams, to guarantee the proper functionality of the retention device and to maximise the productivity with a minimal amount of media consumption. To achieve these goals, solid process development and operation strategies are required.

1.3 DEVELOPMENT AND OPERATION OF PERFUSION CULTIVATIONS

The development of a perfusion process consists in the design of all the operating conditions at which the process has to be run. Some of these are not significantly impacted by the continuous operation mode. This is the case, for example, of the bioreactor environment, which includes temperature, pH, mixing, oxygen concentration, etc., where the same concepts as for batch operation can be applied. However, there are other variables, such as the various flow streams (media feed, harvest, bleed), that need to be properly defined and controlled to achieve and maintain a steady-state operation. The cell retention device needs to be carefully designed to assure stable operation of the bioreactor for the entire duration of the process. Also, media design and cell line selection need to be adapted with respect to batch operation in order to better fulfil the requirements of the continuous operation. These aspects are briefly discussed in the following section in order to draw a global picture of the complex task of designing perfusion bioreactors across the various relevant scales, while a detailed and quantitative analysis is presented in the rest of the book, and particularly in Chapter 4. On the other hand, the different cultivation scales, including scale-down models, that are used to define these operating conditions are described in Chapter 3.

1.3.1 Cell Retention Device

A variety of different cell retention systems has been described in the literature, which are based on differences in either size or density to separate cells or microcarriers from the liquid harvest flow (Bielser et al. 2018, Pollock et al. 2013, Voisard et al. 2003). In fixed-bed bioreactors, the cells are attached to some packed carriers which are kept inside the reactor. Microcarriers are instead constituted by small beads dispersed in the suspension and kept inside the bioreactor by appropriate spin filters or settlers.

For cells in suspension, the separation is more difficult because of their reduced size and density differences compared to the microcarrier beads. Partial retention can be achieved with well-designed gravimetric settlers, but the separation yield decreases as the cell density increases and more performant systems are typically required. Tangential filtration is based on small pore sizes that retain mammalian cells and a tangential flow to alleviate membrane fouling. Various commercial solutions based on this principle exist today and are discussed in Chapter 2.

1.3.2 Process Monitoring and Control

In order to properly operate a perfusion bioreactor, we need to control three liquid flowrates: the media inlet flow (Q_{feed}), also referred to as the perfusion rate; the

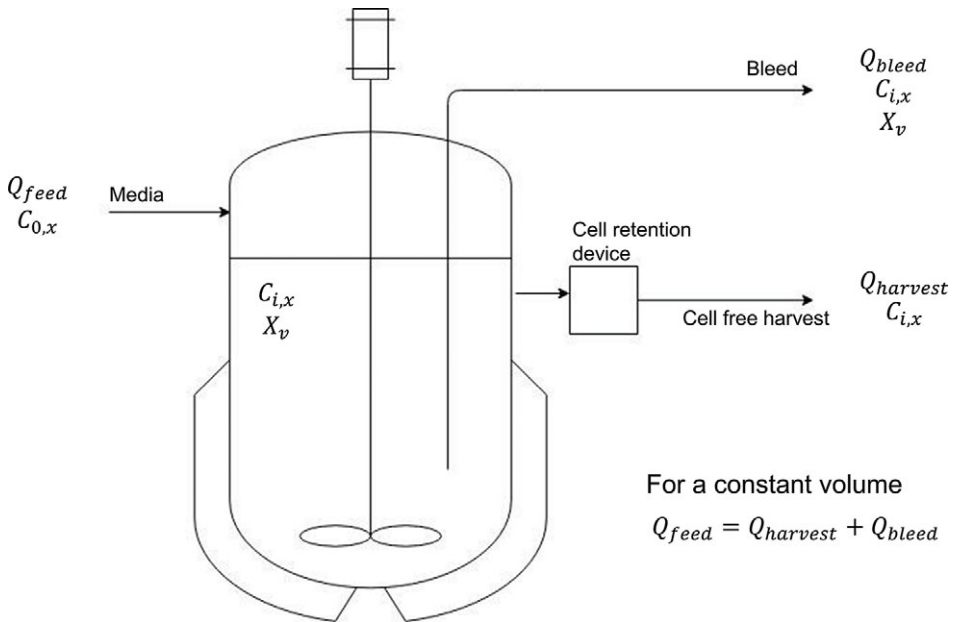


Figure 1.6 Perfusion bioreactor with media feed, harvest and bleed streams. A cell retention device is present on the harvest line. © Reprinted from *Biotechnology Advances*, 36/4, Jean-Marc Bielser, Moritz Wolf, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, ‘Perfusion mammalian cell culture for recombinant protein manufacturing: A critical review’, 2018, with permission from Elsevier.

harvest flow ($Q_{harvest}$) and the bleed flow (Q_{bleed}), as shown in Figure 1.6. In order to maintain a constant volume in the vessel, the inlet flowrate must obviously be equal to the sum of the two outlet ones. These flows can be controlled using different strategies; for example, one can fix either the feed or the harvest flow as being constant and adjust the other two accordingly. The bleed flowrate is typically manipulated in order to control the cell density, while the remaining flowrate – i.e., feed or harvest – is manipulated in order to keep constant the weight of the bioreactor and, consequently, the culture volume.

In order to keep constant or control the flowrate of a liquid stream, one can envision different approaches such as to use a calibrated pump, a non-invasive flow meter based on ultrasounds or a gravimetric flow controller. All options are viable and mostly depend on available equipment. The easiest to implement is, of course, a pump with a defined flowrate, such as a volumetric pump, but this might not be the most precise or reproducible. Therefore, flowrate controllers using a feedback loop based on a measurable physical parameter such as weight or liquid velocity are more recommendable.

The bleed stream is ideally continuous and can be controlled offline, using a pump with a daily flowrate adjustment based on an external cell count, or online, using a capacitance probe that is able to monitor the biomass. A close control of the cell density in the bioreactor is crucial to maintain a stable steady-state operation.

Thus summarising, three streams and one retention device are the extra requirements for perfusion cell cultures compared to fed-batch operations. Technologies are available for their realisation across various scales, from benchtop to manufacturing,

1.3 Development and Operation of Perfusion Cultivations

Table 1.3 Non-exhaustive list of small scale cell culture devices, with comparison of control strategies and operation modes. © Reprinted from *Biotechnology Advances*, 36/4, Jean-Marc Bielser, Moritz Wolf, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, ‘Perfusion mammalian cell culture for recombinant protein manufacturing: A critical review’, 2018, with permission from Elsevier.

Parameters	Deepwell plate	Shake tubes	Microscale	Lab-scale
Mixing	Orbital shaking	Orbital shaking	Stirred	Stirred
pH control	No	No	Yes	Yes
DO control	No	No	Yes	Yes
pCO ₂ control	No	No	Yes	Yes
Aeration	Passive	Passive	On demand	On demand
Cell retention	Centrifugation or Sedimentation	Centrifugation or Sedimentation	Sedimentation	ATF, TFF or other
Perfusion operation mode	Semi-continuous	Semi-continuous	Semi-continuous	Continuous

but the use of cell retention devices is not always straightforward. In particular, the smallest scale may be challenging and requires proper scale-down models.

1.3.3 Scale-Down Models

It is good practice in engineering to develop processes at a small scale and, once the preferred operating conditions are identified, to transfer them to the commercial scale. This is especially true for biopharma processes due to the very high cost and time requirement of any experimental activity. The problem is that continuous flows and cell retention, which are essential features of perfusion cell culture, are difficult to scale down for technical reasons. Batch-like technologies benefit from established scale-down models such as the deepwell plate (DWP), the shake tube (ST) or the micro-scale bioreactor (msBR). Actions like sampling for offline measurements, also referred to as in-process control (IPC), and feeding can, in fact, be easily mimicked in these systems. Strategies for pH and dissolved oxygen (DO) control have also been established using, for example, incubation in a controlled environment and agitation to create mixing and promote interphase oxygen and carbon dioxide transfer in the culture. The characteristics of the various devices available on the market are summarised in Table 1.3.

As mentioned earlier, the case of perfusion is more complex, and this can be mimicked using semi-continuous operations. For this, cells are periodically separated from the fermentation suspension either by centrifugation or sedimentation. The cell-free medium is then harvested and replaced by fresh medium, as shown in Figure 1.7.

Still in the area of scale-down models, but for a bit larger volumes, a device that enables to control up to 24 bioreactors with a culture volume of 250 mL each has been recently commercialised (Zoro & Tait 2017). This device allows the

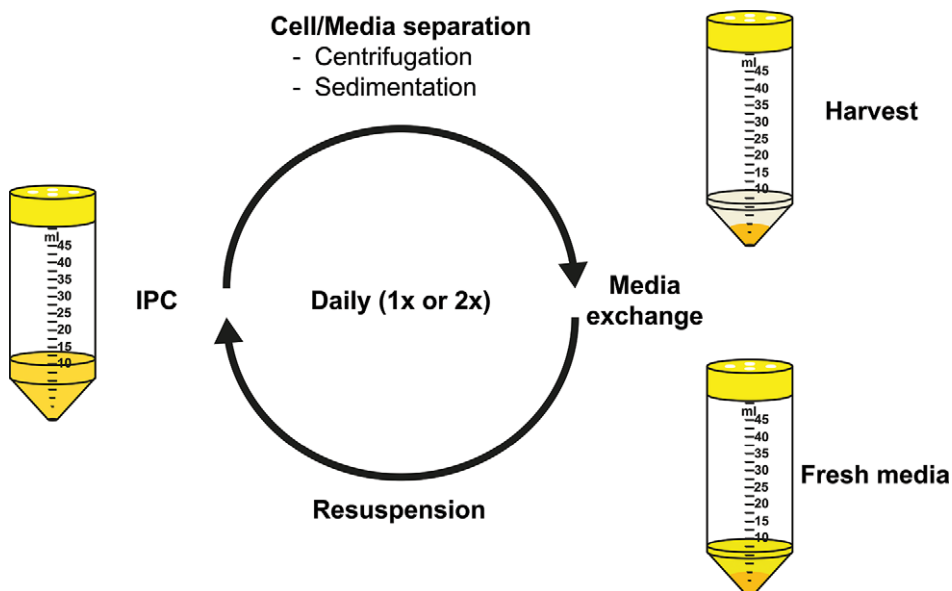


Figure 1.7 Semi-continuous cell cultivation strategy illustrated here for shake tubes. Cells are separated using either centrifugation or sedimentation. Then, the spent media is exchanged with a defined volume of fresh media, and the cells are resuspended. © Reprinted from *Biotechnology Progress*, Jean-Marc Bielser, Jakub Domaradzki, Jonathan Souquet, Hervé Broly and Massimo Morbidelli, ‘Semi-continuous scale-down models for clone and operating parameter screening in perfusion bioreactors’, 2019, with permission from John Wiley and Sons.

implementation of both continuous operation and cell retention. The system offers two filtration patterns: tangential flow filtration (TFF) and alternating tangential flow (ATF) filtration, in addition to online control of pH and DO, at-line control of cell density, and constant media and harvest flow control. There is also a single-use equipment provider (PerfuseCell ©) that has a large choice of different lab-scale bioreactor set-ups, including different types of cell retention devices. This allows for customer-designed bioreactors – including vessel size, number of lines, agitator shape, sparger and even different types of single-use and pre-sterilised retention devices. Another report mentions the development of perfusion capacities at 250 mL scale using modified bioreactor systems (Chotteau 2017). Although more similar equipment is expected to become available soon on the market, those based on semi-continuous operation will remain a valid option due to their low cost and ease of operation. The different characteristics of these approaches and the reliability of scale-down models in the sub-litre range are discussed in Chapters 3 and 4.

I.3.4 Media and Cell Line Development

Ad-hoc media development is a crucial aspect for perfusion processes since it represents a strong component of the production cost. Due to the outlet flow, significant amounts of nutrients, in fact, leave the bioreactor non-utilised. Consequently, the amount of media consumed per mass of product tends to be larger than in batch-type processes. This is a crucial point in the development of perfusion bioreactors that will be discussed in Chapters 2 and 4.

1.3 Development and Operation of Perfusion Cultivations

Another key element of recombinant technology is the development of appropriate cell lines. As mentioned earlier, this is based on the selection of a host cell line and a defined DNA sequence encoding for the protein of interest, which we assume given at this point. For mammalian cells, transient and stable transfection of the given DNA can be performed. Transient transfection is very fast, and the DNA is not integrated into the host genome and, therefore, is not preserved, or copied, over subsequent generations. Nevertheless, for a short period of time the strand will be expressed. This production method is typically used in drug discovery and in the very first development stages when low amounts of product are required and the quality constraints are rather loose (Pham et al. 2006).

However, in order to establish a manufacturing cell line, the transfection must be stable, and the clonality of the cell line must be proven. Different transfection methods exist with mostly random integration (location and number of copies are not controlled) of the vector into the genome. This results in a very large pool of different clones. For transient expression, this pool would be used as such, but for stable transfection, some selection pressure has to be applied in order to identify the rare clones that integrated the transfected DNA into their genome, and thus replicated it with each cell division. The next step is to split these cells individually. Different methods ranging from limiting dilutions to more complex technologies using flow cytometry are used for this separation. Well plates are commonly used to explore a large number of conditions. The screening procedure is based on the clone growth rate and productivity. These are used as indicators to select a number of them, scale them up and repeat the screening procedure with a narrower range of clones. This procedure is repeated until converging to a single clone which is then selected for manufacturing as schematically illustrated in Figure 1.8 (Wurm 2004).

It is to be noted that media design and clone screening occur somehow in parallel and, as the candidates become fewer, the experiments are conducted using larger volumes, thus moving from static plates to agitated deepwell plates, shake tubes, micro-scale bioreactors, lab-scale bioreactors and finally to the pilot and commercial scale (Böhm et al. 2004, Rouiller et al. 2016). Since it is expected that the best clones for operating in the batch mode are not necessarily the best ones in the perfusion mode, the scale-down models used in this procedure should be selected accordingly.

1.3.5 Scale-Up to Clinical and Commercial Production

The process development discussed in the previous sections ranges from the millilitre to the litre scale, which is the one typical of benchtop bioreactors. The next step is to scale the bioreactor up to the clinical and commercial scale.

When considering a traditional stainless steel three-phase bioreactor, this scale-up procedure involves the usual operating variables which are relevant in determining the performance of these reactors such as temperature, mixing, oxygen and carbon dioxide concentration and others, which are rather classical in the operation of three-phase reactors. A wide variety of methods, procedures and models are available for this in chemical reaction engineering and can be easily adapted to the case of bioreactors (Benz 2011, Shah 1979). This is discussed in Chapter 5, where we specifically focus on those aspects which are peculiar of continuous cell cultures. One of the major issues is related to the cell retention device. It is important to extract information from small-scale experiments, such as the maximum flowrate per surface area or the residence time of the cells inside the filtration device, for the appropriate

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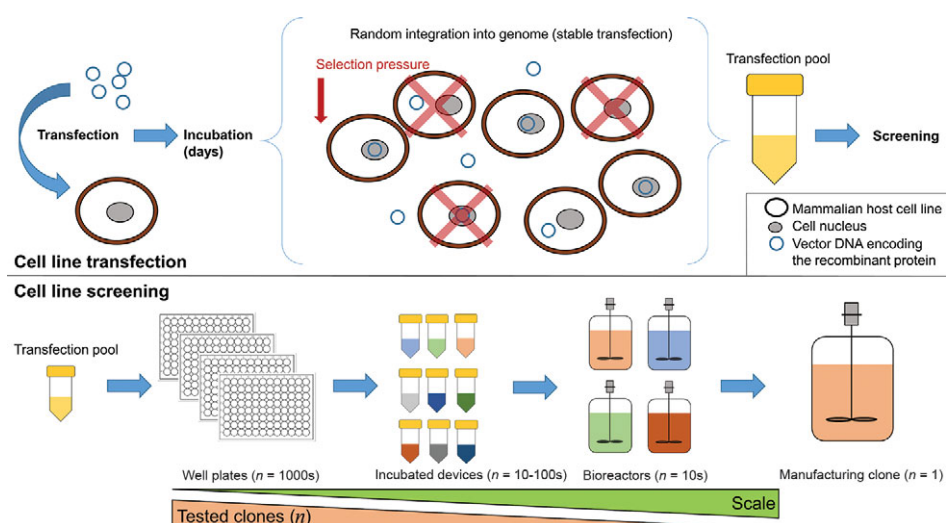


Figure 1.8 Manufacturing cell line development workflow. The cell line transfection step is a random insertion of vector DNA containing the encoding region for the target recombinant protein. Selection pressure is applied to the cell pool in order to favour the survival rate of the cells that integrated the vector. This heterogeneous pool of cells is cultivated for a short amount of time until viability recovers. The second step is the cell line screening. First, the cells from the transfection pool are split into well plates (target is one cell per well). This single cell by dividing gives rise to a clonal population. Each clone is tested in different scale-down platforms (see more details in Chapter 3). As the experimental scale increases, the number of clones decreases until the final choice of the manufacturing clone (Wurm 2004). Colours in the schematic shake tubes and bioreactors each represent a different clone.

design of large-scale units. The challenge is to ensure reliable and robust operation over the entire culture period, which implies sufficient cell retention with negligible product retention, low membrane fouling and appropriate average residence time of the cells in the external loop outside the bioreactor.

Another important trend which needs to be mentioned refers to single-use bioreactors. These technologies have a great potential with respect to decrease the investment costs and simplify unit operations, particularly with respect to cleaning and regeneration operations (Jacquemart et al. 2016). Indeed, several industrial operators are proceeding in this direction and more and more single-use units are becoming available on the market. Nevertheless, these systems are less well characterised than traditional stainless-steel vessels and questions regarding mixing and aeration are often difficult to answer properly. It is clear that we will see more development in this direction in the near future, although it is difficult at this point to predict to which extent they will eventually replace the traditional stainless-steel equipment (Sawyer et al. 2017a).

Finally, Chapter 6 discusses various mathematical models that can be used to describe different phenomenon. Classical models used in reaction engineering can be used to describe mass transport and chemical transformations occurring inside the bioreactor (Shah 1979). Computational fluid dynamics (CFD) is typically used to understand how the agitation impacts the homogeneity of the culture. This is an important feature that needs to be well described in lab-scale bioreactors and, more importantly, at the manufacturing scale. Other types of models can be used to

1.4 Conclusion

describe the cell biology. For example, some metabolic pathways or some particular portion of the cell, such as the Golgi apparatus, can be modelled to understand the impact of changes in the reactor operating conditions. Each one of these – and, even more so, a proper combination of all of them specific for a given target – can provide an important contribution not only in the scale-up of bioreactors but also at the level of process design and development, as well as in understanding the impact of various operating conditions on specific product quality attributes.

1.4 CONCLUSION

Biopharmaceuticals – that is, biologically active proteins used to treat various types of diseases – represent today a well-established and relevant industrial sector. Manufacturing is a very important component of this industry, due to the relatively large production volumes, the high costs and the pressure coming from time and market needs. Regulatory constraints and the lack of understanding/control of the biological component are certainly major factors that refrain sometimes the implementation of the latest technologies. Nevertheless, other drivers like market changes and cost pressure are pushing industry to question its standards and will eventually push for change through innovation.

Perfusion is one of the technologies that could contribute to support such a change. This book is dedicated to perfusion bioreactors and is intended to provide the tools needed to understand their advantages and limitations so as to learn when to use them and how to design them to best exploit their potential across all scales from laboratory to commercial. Although already applied in the past, the current use of perfusion bioreactors in the context of process intensification and continuous manufacturing has a much larger scope which requires a better understanding. It is the scope of this book not only to underline the limitations of this technology but also to indicate the path for their removal or alleviation.

A large part of the book discusses the various aspects of process development of perfusion cell cultures. Some of these are common for all continuous industrial processes, like monitoring and control, while others, such as scale-down models, are particular to biopharmaceuticals. The steady-state concept, which is the basis for the operation of any continuous process, is discussed with some care, due to the presence of living entities which inevitably change their behaviour in time and consequently affect the bioreactor steady-state behaviour.

In the last part of this book, we address the role of large scale perfusion bioreactors in manufacturing biopharmaceuticals and their design and scale-up, also in combination with modelling techniques. This involves the important case of batch process intensification, where perfusion is used coupled to batch-type technologies in order to improve their productivity. On the other hand, perfusion bioreactors can be used alone in the upstream part of the production process, followed by the downstream section where the product is purified to specifications using batch- or continuous-type technologies (Pfister et al. 2018). In the second case, we can aim at an integrated continuous unit, where the up- and downstream are operated in a fully integrated manner, just like a single process. This represents probably the greatest innovation in this area. Unthinkable until a few years ago, it is becoming now reality and an attractive choice in a number of commercial scenarios.